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|---|---|---|---|---|---|---|---|
| **11** | **(a)** | | Component in **i** direction zero $\Rightarrow k = -4$ | **B1**<br>**[1]** | **3.3** | Allow $(-4\mathbf{i} + 5\mathbf{j})$ seen instead of $k$ | Do not allow for $k = -4\mathbf{i}$ or similar. |
| **11** | **(b)** | | Weight $= -0.8g\mathbf{j}$<br><br>N2L<br>$5\mathbf{j} + 3\mathbf{j} - 0.8g\mathbf{j} = 0.8\mathbf{a}$<br>$[\Rightarrow \mathbf{a} = 0.2\mathbf{j}]$<br>$\mathbf{v} = (4\mathbf{i} + 7\mathbf{j}) + 0.2\mathbf{j} \times 10$<br><br>velocity is $(4\mathbf{i} + 9\mathbf{j})$ m s$^{-1}$ | **B1**<br><br><br>**M1**<br><br>**M1**<br><br>**A1**<br>**[4]** | **1.2**<br><br><br>**1.1a**<br><br>**1.1a**<br><br>**2.5** | Allow if seen<br><br><br>Condone missing weight<br><br>Using their **a** in a vector *suvat* equation(s)<br><br>Must be in correct vector form | <br><br><br><br><br><br>Accept fully correct column vector |
| | | | **Alternative solution**<br>Acceleration is vertical so consider only vertical motion<br>$5 + 3 - 0.8g = 0.8a$<br><br>$[\Rightarrow a = 0.2]$<br><br>$v = u + at = 7 + 0.2 \times 10 = 9$<br><br>So the velocity is $(4\mathbf{i} + 9\mathbf{j})$ m s$^{-1}$ | **M1**<br><br>**B1**<br><br><br><br>**M1**<br><br>**A1**<br>**[4]** | | Applying N2L in 1-dimension<br>Condone missing weight<br>Including the weight (consistent sign convention)<br><br><br><br>Using *suvat* equation(s) in the vertical direction<br>Must be in vector form | <br><br><br><br><br><br>Accept fully correct column vector |